Replication Report: (title not recorded)

OSTI 1559043 | Coverage 7/10, Agreement 7/10

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1 Source paper

- **Title:** (title not recorded)
- **Authors:** Not recorded
- **Year:** n/a
- **Domain:** Not recorded
- **Identifier:** OSTI 1559043

2 Replication objective

The central claim of the paper concerns not recorded. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact).

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper's computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/tie`

Paper-directory README excerpt

OSTI 1559043 — Jaravel et al.\ 2019 DNS of ignition-kernel stabilization in cross-flow

Replication status: v4 (20-run Polaris PeleC ensemble) — **7/10** (coverage 7/10, agreement 7/10). Promoted from v3 6/10.

Latest report

• ‘replication-pelec/report_v4/report.pdf‘

Ensemble summary (20-run Polaris, 2026-04-24)

| | N runs | N ignited | IP | paper IP | |——|——|———|——|———-| | 0.6 | 5 | 0 | 0.00 | 0.00 | | 0.8 | 5 | 0 | 0.00 | 0.20 | | 1.0 | 5 | 5 | 1.00 | 0.65 | | 1.2 | 5 | 5 | 1.00 | 0.90 |

Ignition criterion: ‘T_max(0.9 ms) > 2550 K‘ for complete runs **or** ‘max T_max(t>0.5 ms) > 2700 K‘ for CFL-truncated runs. See ‘replication-pelec/report_v4/report.tex‘ §3.

Per-run details: ‘replication-pelec/polaris_ensemble/summary.json‘.

 # Key corrections vs prior notes

• Aurora job **8449654 never ran** (reservation-blocked). Only 8449560 ran — single =0.6 realisation, 9 checkpoints. Aurora is **not** the ensemble source.

• Polaris job set 7099651–7099670 (20 jobs) is the real ensemble.

 # Directory map

[code block omitted]

 # Known limitations (see report_v4 §5)

1. 128×64×64 grid, no AMR level 1 — turbulence under-resolved 2. 1-ms simulation window (paper goes to ~10 ms) 3. Synthetic (not paper's tabulated) turbulent inflow 4. 3 of 5 =1.2 runs CFL-truncated at 0.25–0.63 ms; classified ignited from elevated late-window T_max 5. Per-plotfile species-max extraction on Polaris debug queue; at report time still queued. Hook in ‘polaris_ensemble/analyze.py‘ consumes the per-run ‘*_timeseries.csv‘ when available to add species figures.

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper's resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- Not recorded.

5 What was missing or incomplete

- Not recorded.

The dominant blocker categories for this paper, mapped onto the meta-paper's 9-category friction taxonomy (§4), are listed in §**??** below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §**??**). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Not recorded.

7 Score and friction-point tagging

- **Coverage:** 7/10 — Not recorded.
- **Agreement:** 7/10 — Not recorded.

Friction points encountered (taxonomy tags):

- **F5**: Force-field / hyperparameter drift (unspecified configuration)

8 Follow-on questions

- Not recorded.

9 Reproducibility pointers

- **Paper directory:** `1559043-ignition-kernel-turbulent`
- **Replication artefacts:**
  - `/Users/stevens/Dropbox/REPLICATE-PROJECT/1559043-ignition-kernel-turbulent/replicatio`
- **Replication-dir hint from JSONL:** `/Users/stevens/Dropbox/REPLICATE-PROJECT/1559043-ignition-`
- **Status:** tier_lift_v2_2026_04_27

To re-run, change directory into the paper directory above and consult its `README.md` for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.